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|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CONTACT<br>INFORMATION | <p>Department of Computer Science and Operations Research<br/>University of Montréal</p> <p>Mila, Quebec AI Institute<br/>6666 St. Urbain, Montréal</p> <p><i>E-mail:</i> <a href="mailto:ioannis@iro.umontreal.ca">ioannis@iro.umontreal.ca</a><br/><i>Web:</i> <a href="http://mitliagkas.github.io">mitliagkas.github.io</a></p>                                                                                                                                                                                          |
| RESEARCH<br>INTERESTS  | <p>Statistical machine learning, optimization, deep learning, game theory, generative models, distributed learning systems.</p>                                                                                                                                                                                                                                                                                                                                                                                              |
| AFFILIATIONS           | <p><b>University of Montréal</b>, Department of Computer Science and Operations Research</p> <p>Associate professor June 2022 -<br/>Assistant professor Sept. 2017 - May 2022</p> <p><b>Mila, Quebec AI Institute</b> September 2017 -</p> <p>Core member, Canada CIFAR AI chair<br/>Member of Mila's Scientific Council</p> <p><b>Google DeepMind, Montréal</b>, Staff Research Scientist (part time) September 2022-</p> <p><b>Archimedes, AI Institute</b> January 2024-</p> <p>Affiliate researcher</p>                  |
| PAST AFFILIATIONS      | <p><b>ElementAI, Montréal</b>, Faculty Fellow (part time) 2018-2020</p> <p><b>Stanford University</b>, Postdoc, Departments of CS and Statistics 2015-2017<br/>Supervised by: Associate prof. <a href="#">Christopher Ré</a>, Adjunct prof. <a href="#">Lester Mackey</a></p>                                                                                                                                                                                                                                                |
| EDUCATION              | <p><b>The University of Texas at Austin</b></p> <p>PhD, ECE department. Awarded in August 2015<br/><b>Advised by:</b> Prof. <a href="#">Constantine Caramanis</a> and Prof. <a href="#">Sriram Vishwanath</a><br/><b>Thesis topic:</b> Resource-Constrained, Scalable Learning</p> <p><b>Technical University of Crete</b>, Chania, Greece</p> <p>MSc., ECE department. 2008 - 2010</p> <p>Diploma, Electronic and Computer Engineering (5 year degree), 2002 - 2008<br/><b>Advisor:</b> Professor Nikos D. Sidiropoulos</p> |

RESEARCH GRANTS During my first seven years of tenure-track work, I was awarded a total of about 90 PhD years in competitive funding (about 2.8 million CAD after subtracting any overhead), to be disbursed over a period of 10 years.

- Google x Mila Research grant, awarded 2025-09 — \$60,000
- Meta collaborative grant, awarded 2024-03 — \$91,000
- CIFAR Canada AI chair, (originally awarded December 2018 for five years): Phase II awarded 2023-04, until 2028 — \$1,050,000
- Google x Mila Research grant, awarded 2023-09 — \$50,000
- Microsoft Research collaborative grant, awarded 2021-01 — \$54,000
- Samsung-Mila collaborative three-year grant, awarded 2020-09 — \$240,000 received (of \$2,400,000 total consortium funding)
- IVADO Postdoctoral Scholarship, for my postdoc K. Ahuja, awarded 2020-12 — \$150,000
- CIFAR Catalyst Grant, in collaboration with Manuela Girotti and Murat Erdogdu (UofT, Vector), awarded 2020-09 — \$0 received (co-applicant; \$50,000 total)
- IVADO Postdoctoral Scholarship, Fellow tier, for my postdoc N. Loizou, awarded 2020-01 — \$180,000
- Microsoft Research collaborative grant, awarded 2019-08 — \$54,000
- NSERC Discovery, awarded 2019-04 (+ competitive accelerator supplement) — \$127,500
- CIFAR Canada AI chair, awarded 2018-04 — \$1,052,500
- Fonds de Recherche du Québec, Nature et technologies, Nouveau Chercheur, awarded 2018-04 — \$50,800
- IVADO startup grant (Apogée), awarded 2017-09 — \$91,875
- Université de Montréal startup grant, awarded 2017-09 — \$70,000

#### DOCTORAL STUDENTS

Ryan D’Orazio (PhD candidate)  
Divyat Mahajan (PhD candidate)  
Zichu Liu (PhD candidate)  
Ahmed (Mehdi) Inane (PhD candidate)  
Mehrab Hamidi (PhD candidate)

Yukti Makhija (PhD student)  
Mahdi Ghaznavi (PhD student, starting January ’27)

PAST STUDENTS  
AND SUPERVISEES

Nicolas Loizou (postdoctoral scholar, 2021-12 → assistant prof. at Johns Hopkins Univ.)

Manuela Girotti (postdoctoral scholar, 2021-07 → assistant prof. at Emory University, Atlanta, GA)

Kartik Ahuja (postdoctoral scholar, 2022-09 → research scientist at FAIR Paris)

Kilian Fatras (postdoctoral scholar, 2024-01 → research scientist at Dreamfold, Montreal)

Alexia Jolicoeur-Martineau (PhD, 2023-01 → research scientist, Microsoft Research)

Reyhane Askari Hemmat (PhD, 2024-05 → Meta, Montreal)

Adam Ibrahim (PhD, 2024-05 → Google DeepMind, Mountain View)

Charles Guille-Escuret (PhD, 2025-01 → MBZUAI Institute of Foundation Models)

Hiroki Naganuma (PhD, 2026-07 → NVIDIA)

Vincent Quirion (MSc, 2024-2026 → Soma Energy)

Mehrnaz Mofakhami (MSc, 2024-08 → research assistant at Mila)

Rémi Piché-Taillefer (MSc, 2021-06 → Microsoft Research, Montréal)

Brady Neal (MSc, 2019-12 → Senior Research Scientist at Dataiku)

Oscar Clivio (research intern, 2024-11 → PhD student, University of Oxford)

Lyle Kim (research intern, 2023-01 → PhD candidate, Rice University)

Vincent Quirion (research intern, 2024-08 → MSc at UdeM/Mila, in progress)

Ange-Clement Akazan (research intern, 2023-12 → MSc student at the African Institute of Mathematical Sciences)

Séb Arnold (research intern, 2018-08 → PhD candidate at USC)

Vinayak Tantia (research intern, 2018-08 → research engineer, FAIR Montréal)

Baptiste Goujaud (research intern, 2020-03 → PhD candidate, École Polytechnique)

Amartya Mitra (research intern, 2020-06 → ML Researcher at Morgan Stanley)

Nicolas Gagné (research intern, 2018-08 → PhD candidate at McGill)

TEACHING

**University of Montreal**

Latest teaching evaluation: 3.7/4.0

*Theoretical principles for deep learning*

**Winter 2027**

*Fundamentals of machine learning (in french)*

**Fall 2026**

*Fundamentals of machine learning (in french)*

**Fall 2025**

*Fundamentals of machine learning (in french)*

**Fall 2024**

*Fundamentals of machine learning*

**Fall 2023**

*Theoretical principles for deep learning*

**Winter 2023**

*Fundamentals of machine learning*

**Fall 2022**

*Theoretical principles for deep learning*

**Winter 2022**

*Fundamentals of machine learning*

**Fall 2021**

|                                                      |                    |
|------------------------------------------------------|--------------------|
| <i>Theoretical principles for deep learning</i>      | <b>Winter 2021</b> |
| <i>Fundamentals of machine learning</i>              | <b>Fall 2020</b>   |
| <i>Theoretical principles for deep learning</i>      | <b>Winter 2020</b> |
| <i>Fundamentals of machine learning</i>              | <b>Fall 2019</b>   |
| <i>Theoretical principles for deep learning</i>      | <b>Winter 2019</b> |
| <i>Fundamentals of machine learning</i>              | <b>Fall 2018</b>   |
| <i>Theoretical principles for deep learning</i>      | <b>Winter 2018</b> |
| <b>The University of Texas at Austin</b>             |                    |
| <i>Teaching Assistant—Information Theory</i>         | <b>Spring 2012</b> |
| <b>Technical University of Crete</b>                 |                    |
| <i>Teaching Assistant—Telecommunication Networks</i> | <b>Fall 2008</b>   |

## PUBLICATIONS

- D. Mahajan, S. Goyal, B. Youbi Idrissi, M. Pezeshki, **I. Mitliagkas**, D. Lopez-Paz, K. Ahuja  
Beyond Multi-Token Prediction: Pretraining LLMs with Future Summaries  
arXiv:2510.14751  
*International Conference on Learning Representations (ICLR)*, 2026
- N. Tsikouras, Y. Pantis, **I. Mitliagkas**, C. Tzamos  
A Derandomization Framework for Structure Discovery: Applications in Neural Networks and Beyond  
arXiv:2510.19382  
*International Conference on Learning Representations (ICLR)*, 2026
- K. Yoshida, Y. Naraki, T. Horie, R. Shimizu, **I. Mitliagkas**, H. Naganuma  
DisTaC: Conditioning Task Vectors via Distillation for Robust Model Merging  
*International Conference on Learning Representations (ICLR)*, 2026
- A.M. Inane, V. Quirion, G.K. Dziugaite, **I. Mitliagkas**  
Unlearning with Asymmetric Sources: Improved Unlearning-Utility Trade-off with Public Data  
arXiv:2605.11170  
*International Conference on Machine Learning (ICML)*, 2026
- H. Naganuma, S. Gupta, Y. Briki, **I. Mitliagkas**, I. Rish, P. Raman, H.J.M. Shi  
Adaptive Batch Sizes Using Non-Euclidean Gradient Noise Scales for Stochastic Sign and Spectral Descent  
arXiv:2602.03001  
*International Conference on Machine Learning (ICML)*, 2026
- H. Naganuma, R. Hataya, K. Yoshida, **I. Mitliagkas**  
An Empirical Study of Pre-trained Model Selection for Out-of-Distribution Generalization and Calibration  
arXiv:2307.08187  
*Transactions on Machine Learning Research (TMLR)*, 2025
- L. Maes, TH. Zhang, A. Jolicoeur-Martineau, **I. Mitliagkas**, D. Scieur  
Understanding Adam Requires Better Rotation Dependent Assumptions  
arXiv:2410.19964  
*Neural Information Processing Systems (NeurIPS)*, 2025

- H. Naganuma, X. Zhang, MC. Yue, **I. Mitliagkas**, PA. Witte, RJ. Hewett  
Pseudo-Asynchronous Local SGD: Robust and Efficient Data-Parallel Training  
arXiv:2504.18454  
*Transactions on Machine Learning Research (TMLR)*, 2025
- D. Mahajan, M. Pezeshki, C. Arnal, **I. Mitliagkas**, K. Ahuja, P. Vincent  
Compositional risk minimization  
arXiv:2410.06303  
*International Conference on Machine Learning (ICML)*, 2025
- D. Beaglehole, **I. Mitliagkas**, A. Agarwala  
Feature learning as alignment: a structural property of gradient descent in non-linear neural networks  
arXiv:2402.05271  
*Transactions on Machine Learning Research (TMLR)*, 2024
- E. Triantafillou, P. Kairouz, F. Pedregosa, J. Hayes, M. Kurmanji, K. Zhao, V. Dumoulin, J.J. Junior, **I. Mitliagkas**, J. Wan, L.S. Hosoya, S. Escalera, G.K. Dziugaite, P. Triantafillou, I. Guyon  
Are we making progress in unlearning? Findings from the first NeurIPS unlearning competition  
arXiv:2406.09073  
*Neural Information Processing Systems (NeurIPS)*, 2024
- R. D’Orazio, D. Vucetic, Z. Liu, J.L. Kim, **I. Mitliagkas**, G. Gidel  
Solving hidden monotone variational inequalities with surrogate losses  
arXiv:2411.05228  
*International Conference on Learning Representations (ICLR)*, 2025
- C. Guille-Escuret, PA. Noël, **I. Mitliagkas**, D. Vazquez, J. Monteiro  
Expecting The Unexpected: Towards Broad Out-Of-Distribution Detection  
arXiv:2308.11480  
*Neural Information Processing Systems (NeurIPS)*, 2024
- C. Guille-Escuret, H. Naganuma, K. Fatras, **I. Mitliagkas**  
No Wrong Turns: The Simple Geometry Of Neural Networks Optimization Paths  
arXiv:2306.11922  
*International Conference on Machine Learning (ICML)*, 2024
- D. Mahajan, **I. Mitliagkas**, B. Neal, V. Syrgkanis  
Empirical Analysis of Model Selection for Heterogeneous Causal Effect Estimation  
arXiv:2211.01939  
*International Conference on Learning Representations (ICLR)*, 2024 [**spotlight presentation**]
- R. Askari Hemmat\*, A. Mitra\*, G. Lajoie, **I. Mitliagkas**  
LEAD: Least-Action Dynamics for Min-Max Optimization  
arXiv:2010.13846  
*International Conference on Learning Representations (ICLR)*, 2024 (invited); *Transactions on Machine Learning Research (TMLR)* [featured]
- S. Lachapelle, D. Mahajan, **I. Mitliagkas**, S. Lacoste-Julien  
Additive Decoders for Latent Variables Identification and Cartesian-Product Extrapolation  
arXiv:2307.02598  
*Neural Information Processing Systems (NeurIPS)*, 2023 [**oral presentation**]

- C.G. Escuret, P. Rodriguez, D. Vazquez, **I. Mitliagkas**, J. Monteiro  
 CADet: Fully Self-Supervised Anomaly Detection With Contrastive Learning  
 arXiv:2210.01742  
*Neural Information Processing Systems (NeurIPS)*, 2023
- T. Salvador, K. Fatras, **I. Mitliagkas**, A. Oberman  
 A Reproducible and Realistic Evaluation of Partial Domain Adaptation Methods  
 arXiv:2210.01210  
*Transactions on Machine Learning Research (TMLR)*, 2023
- R. D’Orazio, N. Loizou, I. Laradji, **I. Mitliagkas**  
 Stochastic Mirror Descent: Convergence Analysis and Adaptive Variants via the Mirror Stochastic Polyak Stepsize  
 arXiv:2110.15412  
*Transactions on Machine Learning Research (TMLR)*, 2023
- S. Lachapelle, T. Deleu, D. Mahajan, **I. Mitliagkas**, Y. Bengio, S. Lacoste-Julien, Q. Bertrand  
 Synergies Between Disentanglement and Sparsity: a Multi-Task Learning Perspective  
 arXiv:2211.14666  
*International Conference on Machine Learning (ICML)*, 2023
- H. Naganuma, K. Ahuja, S. Takagi, T. Motokawa, R. Yokota, K. Ishikawa, I. Sato, **I. Mitliagkas**  
 Empirical Study on Optimizer Selection for Out-of-Distribution Generalization  
 arXiv:2211.08583  
*Transactions on Machine Learning Research (TMLR)*, 2023
- S. Sokota, R. D’Orazio, J. Z. Kolter, N. Loizou, M. Lanctot, **I. Mitliagkas**, N. Brown, C. Kroer  
 A Unified Approach to Reinforcement Learning, Quantal Response Equilibria, and Two-Player Zero-Sum Games  
 arXiv:2206.05825  
*International Conference on Learning Representations (ICLR)*, 2023
- A. Mousavi-Hosseini, S. Park, M. Girotti, **I. Mitliagkas**, M. A. Erdogdu  
 Neural Networks Efficiently Learn Low-Dimensional Representations with SGD  
 arXiv:2209.14863  
*International Conference on Learning Representations (ICLR)*, 2023
- M. Mofakhami, **I. Mitliagkas**, G. Gidel  
 Performative Prediction with Neural Networks  
 arXiv:2304.06879  
*Artificial Intelligence and Statistics (AISTATS)*, 2023
- C. Guille-Escuret, A. Ibrahim, B. Goujaud, **I. Mitliagkas**  
 Gradient Descent Is Optimal Under Lower Restricted Secant Inequality And Upper Error Bound  
 arXiv:2203.00342  
*Neural Information Processing Systems (NeurIPS)*, 2022
- K. Fatras, H. Naganuma, **I. Mitliagkas**  
 Optimal transport meets noisy label robust loss and MixUp regularization for domain adaptation  
 arXiv:2206.11180  
*Conference on Lifelong Learning Agents (CoLLAs)*, 2022

- K. Ahuja, D. Mahajan, V. Syrgkanis, **I. Mitliagkas**  
Towards efficient representation identification in supervised learning  
arXiv:2204.04606  
*Conference on Causal Learning and Reasoning (CLeaR)*, 2022
- N. Loizou, H. Berard, G. Gidel, **I. Mitliagkas**, S. Lacoste-Julien  
Stochastic Gradient Descent-Ascent and Consensus Optimization for Smooth Games:  
Convergence Analysis under Expected Co-coercivity  
arXiv:2107.00052  
*Neural Information Processing Systems (NeurIPS)*, 2021
- K. Ahuja, E. Caballero, D. Zhang, Y. Bengio, **I. Mitliagkas**, I. Rish  
Invariance Principle Meets Information Bottleneck for Out-of-Distribution General-  
ization  
arXiv:2106.06607  
*Neural Information Processing Systems (NeurIPS)*, 2021 [**spotlight presentation**]
- A. Jolicoeur-Martineau, R. Piche-Taillefer, **I. Mitliagkas**, R. Tachet des Combes  
Adversarial score matching and improved sampling for image generation  
arXiv:2009.05475  
*International Conference on Learning Representations (ICLR)*, 2021
- C. Guille-Escuret\*, B. Goujaud\*, M. Girotti, **I. Mitliagkas**  
A Study of Condition Numbers for First-Order Optimization  
arXiv:2012.05782  
*Artificial Intelligence and Statistics (AISTATS)*, 2021 [best student paper award OPT2020]
- G. K. Dziugaite, A. Drouin, B. Neal, N. Rajkumar, E. Caballero, L. Wang, **I. Mitliagkas**,  
D. Roy  
In search of robust measures of generalization  
arXiv:2010.11924  
*Neural Information Processing Systems (NeurIPS)*, 2020
- N. Loizou, H. Berard, A. Jolicoeur-Martineau, P. Vincent, S. Lacoste-Julien, **I. Mitliagkas**  
Stochastic Hamiltonian Gradient Methods for Smooth Games  
arXiv:2007.04202  
*International Conference on Machine Learning (ICML)*, 2020
- A. Ibrahim, W. Azizian, G. Gidel, **I. Mitliagkas**  
Linear Lower Bounds and Conditioning of Differentiable Games  
arXiv:1906.07300  
*International Conference on Machine Learning (ICML)*, 2020
- W. Azizian, D. Scieur, **I. Mitliagkas**, S. Lacoste-Julien, G. Gidel  
Accelerating Smooth Games by Manipulating Spectral Shapes  
arXiv:2001.00602  
*Artificial Intelligence and Statistics (AISTATS)*, 2020
- W. Azizian, **I. Mitliagkas**, S. Lacoste-Julien, G. Gidel  
A Tight and Unified Analysis of Gradient-Based Methods for a Whole Spectrum of  
Differentiable Games  
arXiv:1906.05945  
*Artificial Intelligence and Statistics (AISTATS)*, 2020

S. M. Arnold, P. A. Manzagol, R. Babanezhad, **I. Mitliagkas**, N. L. Roux  
 Reducing the variance in online optimization by transporting past gradients  
 arXiv:1906.03532  
*Neural Information Processing Systems (NeurIPS)*, 2019 [**spotlight presentation**]

I. Albuquerque, J. Monteiro, T. Doan, B. Considine, T. Falk, **I. Mitliagkas**  
 Multi-objective training of Generative Adversarial Networks  
 arXiv:1901.08680  
*International Conference on Machine Learning (ICML)*, 2019

V. Verma, A. Lamb, C. Beckham, A. Najafi, **I. Mitliagkas**, A. Courville, D. Lopez-Paz, Y. Bengio  
 Manifold Mixup: Better Representations by Interpolating Hidden States  
 arXiv:1806.05236  
*International Conference on Machine Learning (ICML)*, 2019

A. Lamb, J. Binas, A. Goyal, S. Subramanian, **I. Mitliagkas**, Y. Bengio, M. Mozer  
 State-Reification Networks: Improving Generalization by Modeling the Distribution of Hidden Representations  
 arXiv:1905.11382  
*International Conference on Machine Learning (ICML)*, 2019 [**oral presentation**]

B. Kanuparthi, D. Arpit, G. Kerg, N. R. Ke, **I. Mitliagkas**, Y. Bengio  
 h-detach: Modifying the LSTM Gradient Towards Better Optimization  
 arXiv:1810.03023  
*International Conference on Learning Representations (ICLR)*, 2019

G. Gidel, R. Askari, M. Pezeshki, G. Huang, S. Lacoste-Julien, **I. Mitliagkas**  
 Negative Momentum for Improved Game Dynamics  
 arXiv:1807.04740  
*Artificial Intelligence and Statistics (AISTATS)*, 2019

J. Zhang, **I. Mitliagkas**  
 YellowFin and the Art of Momentum Tuning  
 arXiv:1706.03471  
*Systems and Machine Learning (SysML)*, 2019

P. Achlioptas, O. Diamanti, **I. Mitliagkas**, L. Guibas  
 Learning Representations and Generative Models for 3D Point Clouds  
 arXiv:1707.02392  
*International Conference on Machine Learning (ICML)*, 2018

J. Zhang, **I. Mitliagkas**  
 YellowFin: Adaptive optimization for (A)synchronous systems  
*Systems and Machine Learning (SysML)*, 2018 [**oral presentation**]

C. De Sa, B. He, **I. Mitliagkas**, C. Ré, P. Xu  
 Accelerated stochastic power iteration  
 arXiv:1707.02670  
*Artificial Intelligence and Statistics (AISTATS)*, 2018

T. Kurth, J. Zhang, N. Satish, **I. Mitliagkas**, E. Racah, M.A. Patwary, T. Malas, N. Sundaram, W. Bhimji, M. Smorkalov, J. Deslippe, M. Shiryayev, S. Sridharan, P. Dubey  
Deep Learning at 15PF: Supervised and Semi-Supervised Classification for Scientific Data

arXiv:1708.05256

*Supercomputing (SC)*, 2017

**I. Mitliagkas**, L. Mackey

Improving Gibbs Sampler Scan Quality with DoGS

arXiv:1707.05807

*International Conference on Machine Learning (ICML)*, 2017

**I. Mitliagkas**, C. Zhang, S. Hadjis, C. Ré

Asynchrony begets Momentum, with an Application to Deep Learning

arXiv:1605.09774

*Allerton Conference on Communication, Control, and Computing (Allerton)*, 2016

B. He, C. De Sa, **I. Mitliagkas**, C. Ré

Scan Order in Gibbs Sampling: Models in Which it Matters and Bounds on How Much

arXiv:1606.03432

*Neural Information Processing Systems (NeurIPS)*, 2016

**I. Mitliagkas**, M. Borokhovich, A. Dimakis, C. Caramanis

FrogWild! – Fast PageRank Approximations on Graph Engines

arXiv:1502.04281

*International Conference on Very Large Data Bases (VLDB)*, 2015

D. Papailiopoulos, **I. Mitliagkas**, A. Dimakis, C. Caramanis

Finding dense subgraphs through low-rank approximations

*International Conference on Machine Learning (ICML)*, 2014

**I. Mitliagkas**, C. Caramanis, P. Jain

Memory-limited Streaming PCA

arXiv:1307.0032

*Neural Information Processing Systems (NeurIPS)*, 2013

**I. Mitliagkas**, A. Gopalan, C. Caramanis, S. Vishwanath

User Rankings from Comparisons: Learning Permutations in High Dimensions

*Allerton Conference on Communication, Control, and Computing (Allerton)*, 2011

**I. Mitliagkas**, N. D. Sidiropoulos, A. Swami

Joint Power and Admission Control for Ad-hoc and Cognitive Underlay Networks: Convex Approximation and Distributed Implementation

*IEEE Transactions on Wireless Communications (TWC)*, 2011

**I. Mitliagkas**, S. Vishwanath

Strong Information-Theoretic Limits for Source/Model Recovery

*Allerton Conference on Communication, Control, and Computing (Allerton)*, 2010

**I. Mitliagkas**, N. D. Sidiropoulos, A. Swami

Distributed Joint Power and Admission Control for Ad-hoc and Cognitive Underlay Networks

*International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2010

**I. Mitliagkas**, N. D. Sidiropoulos, A. Swami  
Convex Approximation-based Joint Power and Admission Control for Cognitive Underlay Networks  
*International Wireless Communications and Mobile Computing Conference (IWCMC)*, 2008

WORKSHOP PAPERS

T. Nakamori, L. Gomezjurado Gonzalez, G. Talluri, A. Tiwari, H. Kawashima, **I. Mitliagkas**, G. Rabusseau, H. Naganuma  
Orth-Dion: Eliminating Geometric Mismatch in Distributed Low-Rank Spectral Optimization  
*ICML 2026 CoLoRAI Workshop, arXiv preprint arXiv:2605.16341*

H. Naganuma, J.L. Kim, A. Kyrillidis, **I. Mitliagkas**  
Smoothness-adaptive sharpness-aware minimization for finding flatter minima  
*5th Workshop on practical ML for limited/low resource settings*

M. Mofakhami, R. Bayat, **I. Mitliagkas**, J. Monteiro, V. Zantedeschi  
Performance Control in Early Exiting to Deploy Large Models at the Same Cost of Smaller Ones  
*ICML 2024 workshop on Efficient Systems for Foundational Models, arXiv preprint arXiv:2412.19325*

B. Neal, **I. Mitliagkas**  
In Support of Over-Parametrization in Deep Reinforcement Learning: an Empirical Study  
*ICML 2019 Workshop on Identifying and Understanding Deep Learning Phenomena*

B. Neal, S. Mittal, A. Baratin, V. Tantia, M. Scicluna, S. Lacoste-Julien, **I. Mitliagkas**  
A Modern Take on the Bias-Variance Tradeoff in Neural Networks  
*ICML 2019 Workshop on Identifying and Understanding Deep Learning Phenomena, arXiv preprint arXiv:1810.08591*

J. Zhang, C. De Sa, **I. Mitliagkas**, C. Ré  
Parallel SGD: When does averaging help?  
*Optimization Methods for the Next Generation of Machine Learning Workshop, ICML 2016, arXiv preprint arXiv:1606.07365*

S. Dufort-Labbé, M. Hamidi, R. Pascanu, **I. Mitliagkas**, D. Scieur, A. Baratin  
Navigating Potholes with Geometry-Aware Sharpness Minimization  
*arXiv preprint arXiv:2605.16134*

S. Nakai, Y. Fadhloun, K. Mathlouthi, K. Yoshida, G. Talluri, **I. Mitliagkas**, H. Naganuma  
Generalization Measures under Controlled Covariate Shift: A Regime-Aware Benchmark  
*arXiv preprint arXiv:2602.01718*

O. Clivio, D. Mahajan, P. Taslakian, S. Magliacane, **I. Mitliagkas**  
Learning to Defer for Causal Discovery with Imperfect Experts  
*arXiv preprint arXiv:2502.13132*

AC. Akazan, A. Jolicoeur-Martineau, **I. Mitliagkas**  
Generating Tabular Data Using Heterogeneous Sequential Feature Forest Flow Matching  
*arXiv preprint arXiv:2410.15516*

A. Ibrahim, C. Guille-Escuret, **I. Mitliagkas**, I. Rish, D. Krueger, P. Bashivan  
Towards Out-of-Distribution Adversarial Robustness  
*arXiv preprint arXiv:2210.03150*

M. Girotti, **I. Mitliagkas**, G. Gidel  
Convergence Analysis and Implicit Regularization of Feedback Alignment for Deep Linear Networks  
*arXiv preprint arXiv:2110.10815*

A. Jolicoeur-Martineau, K. Li\*, R. Piché-Taillefer\*, T. Kachman\*, **I. Mitliagkas**  
Gotta Go Fast When Generating Data with Score-Based Models  
*arXiv preprint arXiv:2105.14080*

I. Albuquerque, J. Monteiro, T. Falk, **I. Mitliagkas**  
Generalizing to unseen domains via distribution matching  
*arXiv preprint arXiv:1911.00804*

A. Jolicoeur-Martineau, **I. Mitliagkas**  
Gradient penalty from a maximum margin perspective  
*arXiv preprint arXiv:1910.06922*

A. Ratner, D. Alistarh, G. Alonso, D. G. Andersen, P. Bailis, S. Bird, N. Carlini, B. Catanzaro, J. Chayes, E. Chung, B. Dally, J. Dean, I. S. Dhillon, A. Dimakis, P. Dubey, C. Elkan, G. Fursin, G. R. Ganger, L. Getoor, P. B. Gibbons, G. A. Gibson, J. E. Gonzalez, J. Gottschlich, S. Han, K. Hazelwood, F. Huang, M. Jaggi, K. Jamieson, M. I. Jordan, G. Joshi, R. Khalaf, J. Knight, J. Konečný, T. Kraska, A. Kumar, A. Kyrillidis, A. Lakshmiratan, J. Li, S. Madden, H. B. McMahan, E. Meijer, **I. Mitliagkas**, R. Monga, D. Murray, K. Olukotun, D. Papailiopoulos, G. Pekhimenko, T. Rekatsinas, A. Rostamizadeh, C. Ré, C. De Sa, H. Sedghi, S. Sen, V. Smith, A. Smola, D. Song, E. Sparks, I. Stoica, V. Sze, M. Udell, J. Vanschoren, S. Venkataraman, R. Vinayak, M. Weimer, A. G. Wilson, E. Xing, M. Zaharia, C. Zhang, A. Talwalkar  
MLSys: The New Frontier of Machine Learning Systems  
*White paper, arXiv:1904.03257*

A. Lamb, J. Binas, A. Goyal, D. Serdyuk, S. Subramanian, **I. Mitliagkas**, Y. Bengio  
Fortified Networks: Improving the Robustness of Deep Networks by Modeling the  
Manifold of Hidden Representations  
*arXiv preprint arXiv:1804.02485*

S. Hadjis, C. Zhang, **I. Mitliagkas**, C. Ré  
Omnivore: An Optimizer for Multi-device Deep Learning on CPUs and GPUs  
*Technical report, arXiv:1606.04487*

#### IN THE PRESS

The Greek AI Ecosystem

Trudeau meets with newly appointed Canada CIFAR AI Chairs, **CIFAR News**

Sept chercheurs de l'UdeM titulaires de chaires en intelligence artificielle Canada-  
CIFAR, **UdeM Nouvelles**

NERSC Scales Scientific Deep Learning to 15 Petaflops, **HPC Wire**

De la Grèce à l'UdeM: l'étonnant parcours d'Ioannis Mitliagkas, **UdeM Nouvelles**

#### AWARDS, DISTINCTIONS

Recognized among top reviewers of ICML, 2026

NeurIPS Foundation, top 10% of reviewers, 2020

CIFAR Canada AI chair

NIPS Foundation, listed among best reviewers, 2018

Gerondelis Foundation Inc.: Graduate Scholarship, 2014

The University of Texas at Austin: Microelectronics and Computer Development  
(MCD) Fellowship, 2009-2011

Technical University of Crete: Undergraduate excellence award, 2008

State Scholarships Foundation (Greece): Undergraduate excellence award, 2005

Technical Chamber of Greece: Undergraduate excellence award, 2005

#### PROFESSIONAL SERVICE

Archimedes summer school, Athens, Greece (2022)

Neuromatch Academy Deep Learning Track, World-wide (2021, 2022)

AI4Good Summer School, Canada (2021)

Co-organizer of INFORMS 2021 workshop:  
"Accelerated methods in convex and non-convex optimization"

Co-founder and organizer of [MTL MLOpt](#), a Montreal-based group of researchers from many universities and industrial labs.

Member of the inaugural program committee of MLSys:  
The committee's role was to decide the conference's focus and steer its future goals.

Organizer of NeurIPS 2018, 2019 workshop:  
"Smooth Games Optimization and Machine Learning"

Reviewer of MITACS Accelerate grants

Served as head of the scientific committee in charge of evaluating IVADO grants (2018) and as member in subsequent years.

Host of oral session, NeurIPS 2021

Reviewer and AC for a number of journals and conferences including NeurIPS, ICML, COLT, AISTATS, ALT, AAAI, ICLR, JMLR (editorial board), IJCAI, SIGGRAPH, Transactions on Information Theory, ISIT, ICASSP, Transactions on Wireless Communications.

RECENT INVITED  
TALKS (NOT  
INCLUDING  
ACCEPTED PAPER  
PRESENTATIONS)

|                                                           |                     |
|-----------------------------------------------------------|---------------------|
| Neuromatch Academy Deep Learning Track, World-wide        | Summer 2023         |
| Archimedes summer school, Athens, Greece                  | July 2022           |
| Simons semester in game theory, Berkeley, CA              | January-March, 2022 |
| SIAM conference in optimization OP21, Spokane, WA         | July 2021           |
| Neuromatch Academy Deep Learning Track, World-wide        | Summer 2021         |
| AI4Good Summer School, Canada                             | May 2021            |
| Google Brain, Montreal, QC                                | November 2020       |
| ITA, San Diego, CA                                        | February 2020       |
| INFORMS, Seattle, WA                                      | October 2019        |
| Microsoft Research workshop, Montréal, QC                 | October 2019        |
| Theoretical Advances in Deep Learning, Workshop, Istanbul | July 2019           |
| UT Austin, TX                                             | March 2019          |
| NVIDIA, Webinar                                           | March 2019          |
| ElementAI, Toronto, ON                                    | November 2018       |
| BorealisAI, Toronto, ON                                   | October 2018        |
| USC, Los Angeles, CA                                      | October 2018        |
| Microsoft Research workshop, Montréal, QC                 | October 2018        |
| ElementAI, Toronto, ON                                    | September 2018      |
| Microsoft Research, Montréal, QC                          | August 2018         |
| ElementAI, Montréal, QC                                   | June 2018           |
| FAIR, Montréal, QC                                        | May 2018            |
| RLLab, McGill, Montréal, QC                               | April 2018          |
| ElementAI, Montréal, QC                                   | April 2018          |
| TechAide, Montréal, QC                                    | April 2018          |
| ECE Seminar, UT Austin, TX                                | March 2018          |
| BayesComp, Barcelona Spain                                | March 2018          |
| SysML, Stanford CA                                        | February 2018       |
| Google Brain, Montréal                                    | November 2017       |
| Texas Wireless Summit, Austin, TX                         | October 2017        |
| Colloquium, University of Montréal                        | September 2017      |
| Colloquium, The University of Texas, Austin               | September 2017      |

|                                                                   |                |
|-------------------------------------------------------------------|----------------|
| AutoML workshop, ICML, Sydney                                     | August 2017    |
| Workshop on Advances in Computing Architectures, Stanford SystemX | April 2016     |
| ITA workshop, San Diego, CA                                       | February 2017  |
| AAAI 2017 Workshop on Distributed Machine Learning                | February 2017  |
| Microsoft Research, Cambridge, UK                                 | December 2016  |
| SystemX Stanford Alliance Fall Conference                         | November 2016  |
| Microsoft Research, New England                                   | October 2016   |
| Allerton Conference, Monticello, IL                               | September 2016 |
| Google Brain, Mountain View, CA                                   | August 2016    |
| MIT Lincoln Labs, MA                                              | August 2016    |
| NVIDIA, Santa Clara, CA                                           | July 2016      |